

**Socioeconomic Determinants of International Football Performance: Panel
Data Analysis using an Elo Rating**

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Abstract:

This paper explores the socioeconomic and political determinants of football at an international level, using an Elo rating to investigate all competitions and matches. Developing a theoretical foundation based on literature and theory, a panel data analysis of the years 1994-2018 is employed using ordinary least squares (OLS) and negative binomial regression. Hypotheses are generated based on this framework and original variables are explored. Findings suggest that nations with higher standards of living positively affects the national team's success, the implications of democratic regime and conflict are also explored with mixed results.

Keywords: football success, macro factors, socioeconomic variables

Glossary of Abbreviations:

FIFA = Fédération Internationale de football association

WC = World Cup

AFC = Asian Football Confederation

CAF = Confederation of African Football

CONCACAF = Confederation of North, Central America and Caribbean Association Football

CONMEBOL = South American Football Confederation

OFC = Oceania Football Confederation

UEFA = Union of European Football Associations

GDP = Gross-Domestic Product

HDI = Human Development Index

OLS = Ordinary Least Squares

Centre for Systemic Peace = CSP

United Nations Development Programme = UNDP

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Word Count: 8246

1. Introduction

1.1. Background Information

Football is a global phenomenon, credited as the most popular sport worldwide, it has become a religion for millions, creating delight one day and despair the next.

The governing body of world football, FIFA or the Fédération Internationale de football association has 211 affiliated members (FIFA, 2024). The inaugural FIFA World Cup took place in Uruguay in 1930, hosting a mere 13 teams from the Americas and Europe. Apart from the years encompassed by the second world war, FIFA has organised the global tournament every 4 years, with the latest edition of the tournament, the 2022 Qatar World Cup hosting 32 teams from around the world. Countries (associates) are divided into six confederations based on geographical location, AFC (Asia), CAF (Africa), CONCACAF (Central/North America), CONMEBOL (South America), OFC (Oceania) and UEFA (Europe) and these make up the basis of the international footballing system where countries play regional qualifying matches to determine whether they make it to the World Cup. In addition, regional tournaments are also organised by each confederation, such as the European Football Championships (EUROS, for Europe) or the Copa América for South America.

It can be said that football has engulfed every corner of the globe, in addition to the 211 FIFA associations, over half the world's population at 5 billion engaged with the 2022 Qatar World Cup (FIFA, 2023), with the final reaching an audience of close to 1.5 billion. It's clear that football has become a significant part of globalization and cultural exchange, the 21st century has seen the sport skyrocket into a multi-billion-dollar industry with staggering revenues for clubs, players, institutions, and everybody in between. The financial climate of football has become a key point of investigation, a love of football in many cases involves a love of numbers, the scores, the dates, the league tables, football has quickly involved into a numbers game and thus the determinants of on-field performance are so important (Kuper and Szymanski, 2018). In 2026, the United States, Canada and Mexico will host an unprecedented 48 teams in the World Cup, the format change allows more countries to participate in the WC. In addition to the FIFA World Cup, regional tournaments are quickly becoming just as important, the European Championships, Copa América and African Cup of Nations have significant economic impact and attract a lot of international attention, with the UEFA Euros final in 2020 attracting a cumulative live audience of 5.2 billion people, reaching over 220 different countries (UEFA, 2021). While many studies focus on the World Cup alone, the sparse format of every 4 years limits the accuracy of research, particularly in modern times. Using an Elo

system, this study reflects international football as a whole, weighting different tournaments and matches to properly measure success and performance.

For many countries, football is ingrained in the culture, tradition gives rise to arguably the most important incentive for countries competing internationally: global bragging rights and reputation. Deep rooted footballing tradition exists especially in Latin nations such as Portugal, Argentina and Uruguay, where the sport developed substantially, and institutions were formed outside of Britain (Hoffman, Ging and Ramasamy, 2002b). These deep traditions and cultures create a patriotic pride in representing one's country, giving players an increased incentive to perform at their very best not only in the World Cup but also in regional tournaments as well as friendlies.

Microcosms are often used to demonstrate the effects of economic foundations, Gásquez and Royuela (2014) illustrate that football performance can be used as an instrument of socioeconomic development, giving rise and incentive to study the parallels between football and the determinants of on-field performance. Unlike club football where any given club can hypothetically be purchased by billionaires and surged into success, national teams must wholly rely on their domestic resources and infrastructure, rallying local assets to compete on the largest stage. When it comes to determining success in international football the Olympics is often used as an early model, Bernard and Busse (2000) proposed that population is the most important factor in determining country medal totals, larger countries have deeper talent pools and thus a greater chance at winning. In the case of football, this simply isn't the case, the two largest countries on the planet by far, China and India, are both non-existent when it comes to international football success. As a result, a deeper analysis into the determinants of national performance must be undertaken.

This study examines the political and socioeconomic factors which impact performance in men's international football. While several papers have explored the determinants of performance, the vast majority have used the flawed FIFA rankings or total points to measure performance. This paper uses a modern Elo rating, which utilizes a weighting system to accurately capture the different importance of matches. In addition, the use of the Human Development Index reflects a neglected gap in the literature, where many authors use gross-domestic product as an

overarching proxy for wealth and development. In modern times, HDI has become a strong alternative to GDP per capita, which is often seen as one-dimensional compared to HDI which explores more indicators of development (Sagar and Najam, 1998). HDI contains three indicators: GDP per capita, life expectancy and literacy rates for education. Som (2022) highlights that countries with higher performance in sport generally have higher HDI scores, commenting that sport should be included as an indicator of measurement of HDI. Thus, there is a rationale to analyse the relationship between HDI and international football performance.

Leeds and Leeds (2009) and Jiménez (2016) provoked an academic interest into the impact of political institutions on football performance, arguing and concluding that regimes, colonial heritage and institutions impact a nation's football performance. This study looks to update and expand onto this area of research, using the Polity IV Index to determine a nation's democratic status and incorporating conflict variables, a theoretical framework is modernized and applied to 149 countries from 1994-2018. An OLS regression is used to analyse the relationship between Elo rating, political and socioeconomic variables with a negative binomial regression being used to provide robustness.

1.2. Research Question

To evaluate the political and socioeconomic determinants of international football performance.

1.3. Aim

Critically assess the relationship between Elo rating and socioeconomic variables.

1.4. Objectives

- Evaluate the significance of the Human Development Index and Democratic status as determinants of international football performance.

- Update and expand upon relevant literature by employing a wide panel data approach to 147 countries within 1994-2018, using interaction terms to measure variable dependency and make new inferences.
- Make valuable suggestions for future literature on the strength of using the Elo rating ahead of traditional FIFA rankings.

1.5. Structure

The next section of this paper will review the literature. A theoretical review will discuss the Human Development Index, Democracy and Stability and their uses as macroeconomic determinants. Following this, an empirical analysis will focus on the findings of existing literature with relevant hypotheses surrounding the relationships between the variables. The third section will outline the data, methods, methodology and establish suitable models for this paper following a theoretical framework. Following this, the fourth section will critically evaluate and discuss results of the analysis. Finally, the fifth section will conclude the paper, summarise, and make recommendations for future literature.

2. Literature Review

This literature review is split into two sections, the first section will critically review the theoretical determinants of football performance: including Human Development Index, democracy and conflict. The second section will carry out an empirical analysis of the existing literature.

2.1. Theoretical Review

The Human Development Index is a marker of socioeconomic status and quality of life, based on life expectancy, years of education and per capita income, Santos et al (2019) argues that HDI is the strongest variable to influence a favourable environment for sportive development alongside cultural factors. Given that HDI captures multiple dimensions of development, including the overlap of wealth reflected through per capita GDP, its seen as a strong alternative to GDP (Sagar and Najam, 1998). While GDP is the most frequently used proxy for wealth in studies exploring the determinants of Olympic and international football success, GDP should not serve as a singular basis for human well-being and progress (Brinkman

and Brinkman, 2011). Life expectancy reflects longevity and health, an important factor in fostering the right conditions to train and develop youth talent, in the modern game its paramount that the top level of footballers must look after their body to a ridiculous degree, countries with better health standards can provide the right facilities and care. The educational impacts on football performance have not been properly explored, a logical theory would be that countries with the very highest levels of literacy may see worse international football results because more money is invested into technology and electronics, which pull the focus away from sports. Only Fan et al (2023) has empirically tested the impact of HDI on international football performance, with a limited commentary and discussion, there is a clear rationale to explore the HDI in terms of sporting success.

The impact of wealth and income on international football performance is a highly debated topic amongst researchers, on the one hand, the development of sporting talent is dependent on a nation's sporting infrastructure. Wealthier countries are more likely to outlay expenditure into facilities, equipment, coaching, and talent scouting at different levels (Hoffmann et al, 2002a). Kuper and Szymanski (2018) argue that there is a common myth regarding the link between poverty and producing the best athletes, namely the sentiment that "sport is their only escape from poverty." Using the example of South Africa, the authors argue that the reason they are bad at football is because they didn't give enough good food. At a developmental level, countries which cannot build the fittest and strongest men are argued to be at a disadvantage, which continues into professional sports. However, football is a relatively capital un-intensive sport, unlike other sports, the equipment needed to play, enjoy, and improve is little, as is the amount of organisation that is required (Hoffmann et al, 2002b). This may be the basis as to why Latin countries who were once much poorer developed an increased popularity in football and became a powerhouse of talent production, many top footballers from Brazil for example are from very poor neighbourhoods. In addition, it can be argued that families with more disposable income can afford to purchase electronic forms of entertainment for the children, shifting the interest of youngsters from outdoor to indoor activities (Hoffmann et al, 2002b).

Footballing tradition and culture is a commonly used variable to explain international footballing success, while different proxies are used there is a clear connection

between a long-standing history of football and high levels of performance in the modern game (Fan et al, 2023; Jiménez, 2016; Hoffmann, Ging and Ramasamy, 2002b; Torgler, 2006; Monks and Husch, 2009; Gásquez and Royuela, 2016). In line with Hoffmann Ging and Ramasamy, as well as Monks and Husch, hosting the World Cup is used as a proxy for a footballing tradition. Hosting activity is an indicator of sporting culture, certain countries exhibit a “cultural affinity towards football” (Hoffmann, Ging and Ramasamy:261). For a nation to host the tournament, they must have financial capabilities, a good level of infrastructure but most importantly, public support. These countries have strong, long-standing football institutions which provide a solid foundation for the modern national teams, typically strong domestic leagues like that found in Argentina, Brazil and Europe as well as strong cultural attractions to icons, like the craze associated with footballing legends like Maradona, Pele and more recently, Cristiano Ronaldo and Lionel Messi. It’s theorised that the cultural attraction to the sport will result in better national performance. In addition, years of FIFA association is also used to provide robustness and reflect Torgler (2006) and Gásquez and Royuela (2016).

Democracy generally involves transparent and fair regulations in domestic leagues, including a competitive transfer system which cycles money throughout national institutions and improves the general level of talent as well as adding to competitive market conditions. These factors are generally held to improve football industries nationally, sustainably (Wan et al, 2020). Five out of the eight winners of the World Cup correspond with Europe’s “Big Five Leagues” – Spain, Italy, France, Germany and England, with a combined brand value of over 17 billion euros these countries dominate (Statista, 2023) club football and attract the biggest talents. They are also all historical democracies leading the world in freedom. Because democracies tend to produce stronger domestic leagues, a virtuous cycle is established leading to first class competition, in theory this allows a nation to outperform other regimes internationally (Jiménez, 2016). Despite several autocratic nations investing heavily in football, the regime does not seem to suit the sport as it relies heavily on the collective actions of fans and private businesses, which are argued to be threats to the regime and as a result, national teams under autocratic regimes rarely perform well (Wan et al, 2019). Theoretically however, regime type should not affect the availability of naturally gifted players in a population, both democracies and

autocracies can establish youth academies and programs that generate a certain level of skill (Jiménez, 2016). The argument can be made that autocracies may pose a greater threat in international football, as football is heavily linked to bolstering nationalism and regimes built on collective power may be superior at coming together as one with a common goal; to perform internationally.

Democracy or autocracy alone cannot tell the entire story, while dictatorships are prone to instability during transition periods, their ability to stabilise can cultivate the essential economic conditions to enhance their national performance (Jiménez, 2016). Likewise, a democracy in a constant state of instability may not provide confidence to investors, leading to a worse level of national league and therefore a worse level of international performance. Conflict is greatest in governments in transition and least among established democracies and authoritarian regimes, implying that conflict and instability regularly coexist. Countries involved in conflict are expected to perform worse, resources are pulled away from the institutions that provide national teams with readily available talent and the public focus draws away from leisure and footballing competitions, reducing the “cultural” aspect. Nyanjom (2020) outlines the problems with Kenya’s footballing institutions, arguing that power struggles, political mismanagement and an overall lack of adequate governance has majorly contributed to the problems. On the flip side, countries who stay away from conflict, have superior conditions to allow football institutions to prosper as more investment is available, leading to international success. As resources will not be strained as intensely by themes of instability like conflict, unrest, and poverty, they can be allocated into developing institutions and policies which look to improve and encourage sports, starting at younger levels. Higher involvement in conflict results in lower growth rates in per capita GDP, while countries enduring vulnerable situations may not choose to invest in sporting infrastructure and institutions, financially they physically cannot as there is no spare capital (Aisen and Veiga, 2011). It’s also found that instability also adversely affects productivity and leads to less efficient resource allocation, which negatively affects a country’s organisational abilities in regard to football institutions, which would theoretically lead to poorer national performances (Gásquez and Royuela, 2016). Countries involved in conflict and warfare have a reduced ability to mobilise resources and labour for leisure purposes, including the

development of efficient football institutions which help increase the standard of play in a given nation.

In all professional team sports, the team playing at home has an advantage. Pollard (1986) summarises the aspects which contribute to a positive home advantage, local crowd support stimulates energy for the home team, while intimidating the opposing players, this can also affect the referee's decisions which tend to favour the home team. Travel fatigue has an adverse effect on performance, especially for international football when long distances are involved. Another very important factor is the familiarity of local conditions, such as temperature and altitude, it can be assumed that Brazil and Argentina would have an easier time competing in Africa, where climatic conditions are relatively similar, compared to a Nordic nation like Norway or Sweden (Pollard and Armatas, 2017). While the effects of home advantage differ for club and international football, the geographical impacts tend to be strong, favouring host and nearby nations. While this paper has a broad focus on international football as a whole, and not specifically the World Cup, home advantage is included because the incentives to perform at the World Cup are massive. In addition, hosting the tournament would logically "boost" national performances with a knock-on effect, for example after hosting the 2006 World Cup, Germany reached the finals of the UEFA Euros two years later, finished third in the 2010 World Cup and then won the 2014 World Cup.

2.2. Empirical Analysis

In the 1970's the Olympic Games was to be the lens at which sociologists and economists analysed the socioeconomic impacts on sports. Early consensus hypothesised that population and economic growth were strong determinants of winning medals, as well as political ideology which was a specific focus due to the ongoing Cold War (Ball, 1972; Grimes et al, 1974; Levine, 1974). Until the 21st century, literature struggled to develop past these early publications, Kuper and Sterken (2001) speculate that global conflicts were causing inconsistent attendance for the Olympics, halting research into the subject.

Hoffmann, Ging and Ramasamy (2002b) modernised earlier literature to expand into international football, analysing a cross-sectional study for 76 countries in the year 2000. FIFA rankings are used as the dependent variable due to availability, values

are calculated on a monthly basis using the performance of all matches regardless of importance over the last eight years. Youth development is dependent on the existence of organisational infrastructure and the availability of good quality equipment, in addition private access to resources and leisure time are equally as important, as a result the authors opt to use GDP per capita as a proxy for the influence of these factors. Findings suggest an inverse U-shaped relationship between wealth and performance with an optimal level of per capita wealth at \$21,836, slightly less than the Western average in 2002. Per capita GDP beyond this point results in lower rankings, suggesting that the richest countries may not be the best at football. Torgler's (2006:116) wide panel analysis supports this conclusion, identifying that greater wealth is subject to the "law of diminishing returns." This supports the theoretical viewpoint that wealth shifts the popularity of activities from outdoor to indoor (Hoffmann et al, 2002b). In terms of HDI, Fan et al (2023) found that HDI was statistically significant at the 10% level, with a surprising negative coefficient suggesting that higher levels of development and standard of living correspond to worse international football results. The relationship between sporting success and health, education and productivity were found to be non-direct, which gives us further rationale to build upon this conclusion.

Average annual temperature is a commonly used explanatory variable with Hoffmann et al (2002b) hypothesising that extreme daytime temperatures and excess humidity result in outdoor sports becoming less popular and attractive and therefore reducing the development of talent. Similarly to wealth, temperature depicts a quadratic relationship with 14 degrees Celsius identified as the optimal temperature, deviations from this result in lower FIFA rankings overall. The authors also recognise the importance of the cultural impacts on football, identifying Latin America as a key region of popularity within the sport. An interaction term between Latin origin and population is found to be significant, implying that population size has no impact of performance unless within the Latin world, contradicting Bernard and Busse (2004)'s early model of medal (success) share. While the study is pivotal in establishing a foundation in which further research into the determinants of performance would be built upon, the use of a cross-sectional analysis weakens the results, future studies would improve upon this by considering a wider analysis and building a more accurate picture.

Leeds and Leeds (2009) explored the impact of political and colonial factors, arguing that institutions are the key element for economic growth which is reflected in their production function which emphasises the importance of resource mobilization. Little significance is found between the degree of freedom of a nation and international performance and the cross-sectional nature of the study limits its scope to only 2006, restricting its findings. On the other hand, Jiménez (2016) found that democratization is an underlying factor in the improvement of a team's performance, for every improvement on the democratic scale, a team's winning percentage would increase by 0.33 percent. The study used the PolityIV index to measure political freedom, as well as utilising political stability, measured by the years since the last major political shift. A wide panel data study is employed to update on Leeds and Leeds (2009), with a broad focus on World Cups between 1930 and 2010. Results from the OLS model suggests that there is an interaction between freedom and stability, but there is a clear lack of discussion regarding the theoretical impact of stability or instability, only providing a brief hypothesis. This study looks to build upon this area of research, providing new outlook on how political stability based on conflict impacts performance at the World Cup and all international football competitions.

Results from Torgler (2006) find that a football tradition strongly affects the performance of national teams, using current host as a proxy. Fan et al (2023) confirms these findings, adding that a country that previously hosted the World Cup will find higher levels of success in the future, suggesting that previous host nations contain an ingrained culture of football and tend to outperform countries that have not had the honour of hosting. Findings from Hoffmann, Ging and Ramasamy (2002b) state that world championship hosting has a positive impact on performance both inside and outside the World Cup. It's important to note that when a country hosts the World Cup, they also have the added benefit of a home advantage, Monks and Husch (2009) find that because of culture, lack of travel fatigue and fan support host countries exhibit higher levels of performance. Findings prove the advantage of hosting the World Cup to be statistically insignificant, beyond that of being a "seeded" team and playing in your home continent. This is not consistent with earlier studies from Pollard (1986) and Pollard and Armatas (2017) who explored club football, this may be because international matches allow teams and squads to

acclimatise to their environments, lessening the impact of the “away from home” sentiment.

Additionally, the years a nation has been associated with FIFA is another proxy for tradition which is explored in earlier literature, Houston and Wilson (2002) found a positive but insignificant relationship with Torgler (2006) supporting this conclusion, arguing that nations with longer associations with FIFA typically have rule setting power, leading to a skew in results.

Drawing on both the theoretical and empirical review, a selection of hypotheses can be made regarding the impact of development, wealth, political regime and stability on overall football performance.

H₁: Higher levels of development and wealth will contribute to increased levels of performance.

H₂: Democratic regimes will see better levels of international football performance.

H₃: Countries with greater amounts of violent periods will exhibit poorer overall performance.

3. Methodology

3.1. Data

This study is one of only three to explore the Elo rating as a dependent variable with regard to international football performance, World Football Elo Ratings are based on the original rating system developed for FIDE, the international chess federation (eloratings.net). Ratings look to calculate a team’s true strength relative to its competitors, giving differing weights to tournaments based on global importance, with the World Cup Finals given the most weight and friendly matches having the least. While the most commonly used measurement of international football performance are FIFA rankings, a number of methodical problems have been identified:

1. FIFA ranking only considers a win, loss or draw.

2. The system is not IRB; no differing weight is given depending on opponent weakness/strength.
3. Short time horizon and methodological changes since its inception in the 1990s.

The Elo system solves a number of problems present in the FIFA rankings, the use of a low volatility index reflects more accurate ranking changes, geographical issues are avoided as the confederation a nation belongs to is not incorporated (Gásquez and Royuela, 2016). There are considerations regarding expected and actual goal difference in the game, and nations receive less Elo if they beat a team that they were expected to, likewise “upsets” receive greater amounts of points, Stefani and Pollard (2007) argue that a predictor system (IRB) is highly recommended for those seeking a reasonable and accurate rating system. The natural logarithm of Elo is taken as a response to evidence of heteroscedasticity, also allowing an increased linear relationship between Elo and HDI.

Human Development Index (HDI) data is collected from the United Nations Development Programme, values range between 0 and 1, with the latter being of a higher standard of living. HDI is used as a proxy for both social development and wealth, as it accounts for per capita GDP, as well as life expectancy which can be seen as a proxy for the level of health of a nation, as well as an educational indicator. In line with all papers on the topic, the natural logarithm of population (LPOP) is used to represent a country’s potential pool of talent, since the log of population maximises the R^2 value it’s taken ahead of population share (Gásquez and Royuela, 2016).

For both political variables of interest, democracy and stability, data is collected from the Centre for Systemic Peace, who have developed a number of datasets including the Polity5 Annual Time Series which has information on independent states for the period of 1800-2018. The actual measurement of democracy (DEMOC) is known as “Polity2” where fully autocratic nations take a value of -10 and fully democratic nations take a value of 10, this is used as a proxy for political freedom and regime as detailed data on specific regimes in 2024 are not widely available. While Jiménez (2016) used the years of passage since a major political shift as a proxy for stability, this study opts to use a dummy variable (CONFLICT) which takes the value of 1 if a

nation was subject to any sort of conflict, including international violence, civil war and ethnic violence. The data is also collected from the CSP using the Major Episodes of Political Violence dataset.

Temperature data was the hardest to find, national averages were collected from weatherandclimate.com and are computed on an average across the country. The difference between 14 degrees (the discovered optimal temperature) and the average temperature of the nation is taken and made positive, leading to the TEMP14 variable.

To measure tradition and culture, two variables have been selected in line with literature from Torgler (2006) and Fan et al (2023). Previously hosting a World Cup (PASTHOST) is a dummy variable which takes the value of 1 if a nation has previously hosted the tournament, with a single year lag, for example the United States first hosted the tournament in 1994 but the variable did not take 1 until 1995. Data was collected from Fbref.com, which has detailed information on World Cups. In addition, years of FIFA membership (MEMBER) data was gathered from FIFA.com, measuring how long a nation has been an associate member. Founding members like Germany, France, Spain, Italy and England have the longest memberships, with South American countries getting involved in the organisation not long after its founding.

To measure the impact of a home advantage, a dummy variable is used based on whether a nation has hosted the World Cup (which takes the value 1 until the next tournament, to model any knock on effects hosting might have) or their respective regional tournament (Asian Cup, EUROS, Copa America, AFCON). Data on hosting was taken from Fbref.com.

Table 1 – Variable definitions and their sources

Variable	Definition	Study	Source
LELO	Log of Elo Rating	Gásquez and Royuela (2016)	www.eloratings.net
RANK	Elo Rank (e.g. highest rating in a year receives #1)	Original	www.eloratings.net
HDI	Human Development Index	Wan et al (2016)	United Nations Development Programme
LPOP	Log of Population	Leeds and Leeds (2009)	World Bank
DEMOC	Polity2 Democracy and Autocracy Scale	Jiménez (2016)	Systemic Peace
CONFLICT	Whether or not a country has been involved in an episode of conflict	Original	Systemic Peace
TEMP14	Deviations from 14 degrees Celsius	Hoffmann, Ging and Ramasamy (2002)	Weatherandclimate.com
PASTHOST	Whether or not a country has previously hosted the World Cup	Fan et al (2023)	Fbref.com
MEMBER	Years since country associated with FIFA	Torgler (2006), Gásquez and Royuela (2016)	FIFA.com
HOST	Whether or not a country has hosted the World Cup or a regional tournament	Original ¹	Fbref.com

¹ Monks and Husch (2009) and Torgler (2004) used home advantage as an independent variable but did not incorporate regional tournaments.

Table 2 – Summary Statistics

Variable	Obs	Mean	St. Dev.	Min	Max
ELO	3,650	1476.75	292.98	529	2177
LELO	3,650	7.27	0.22	6.27	7.69
RANK	3,650	73.43	42.15	1	146
HDI	3,564	0.67	0.17	0.23	0.96
POP	3,650	4.39e+07	1.48e+08	401655	1.40e+09
LPOP	3,650	16.23	1.51	12.90	21.06
DEMOC	3,650	3.85	6.28	-10	10
CONFLICT	3,647	0.16	0.36	0	1
TEMP14	3,650	8.50	4.97	0.40	18.04
PASTHOST	3,650	0.09	0.29	0	1
MEMBER	3,650	55.15	28.77	0	114
HOST	3,650	0.17	0.13	0	1

As the summary statistics show, most of the data is easily comparable, with only population reaching numbers that are not analogous. Elo is also difficult to work with, since it's not like a "winning percentage" that can be translated easily, there is however a very large dispersion of data suggesting that national teams differ massively in strength. The average HDI score is 0.67, suggesting that countries in the dataset on average are fairly developed, a minimum value of 0.23 is interesting, which could either suggest that omissions in the data are in the form of low-income countries, or since the turn of the century nations are moving out of the lowest HDI bracket. The average nation is considered more democratic than autocratic, with a mean of 3.85 which is expected especially with data covering a modern period. A standard deviation of 6.28 suggests that countries are not clustered around that mean, many countries are completely autocratic like Bahrain, or completely democratic like Austria. A low conflict average of 0.16 suggests that countries are not witnessing many periods of violence and conflict, which may suggest that on average countries have the right conditions to foster a good level of international football performance. The mean of TEMP14 tells us that on average a countries

temperature deviates by 8.5 degrees Celsius, with the smallest deviation of 0.4 degrees occurring in South Korea, which based on climatic theory should foster the best conditions for professional football development. On average, nations have been associated with FIFA for 55 years, with an obvious high dispersion in the data because some nations are “founding members” whereas some joined in 1994 at the start of the dataset.

3.2. Data Limitations

Out of the 241 nations on the Elo ratings database, this study focuses on a panel of 147 of them. While data for the sport is collected on micro nations and nations that are not fully recognised by international authorities, data on the political and socioeconomic variables does not cover this many countries and therefore a number of omissions were made. It should be noted that a number of changes were made to the database to ease in the analysis, firstly in line with the literature since FIFA recognises the four home nations of the United Kingdom (England, Scotland, Wales and Northern Ireland) whereas for socioeconomic indicators, they are a collective, England is opted to represent the UK as the largest home nation (Gásquez and Royuela, 2016; Jiménez, 2016). Secondly, Serbia is naturally chosen to represent Serbia & Montenegro, who split in 2006 after they gained independence (Fan et al, 2023) as well as Yugoslavia who were present in Elo ratings despite their dissolution until the early 2000s.

3.3. Theoretical Framework

Bernard and Busse (2004) developed a model for determining Olympic success, theorising that a country should win the number of medals based on their population. Three assumptions must take place for this to hold true, firstly medal-calibre athletes would have to be distributed globally in proportion to each country’s population, secondly Olympic medals are assumed to be proportional to athletic talent. Lastly, we assume that there is not a full global attendance, i.e. not every country will take part in the Olympic Games. Adjusting these assumptions for international football, we can apply the model to footballing success, for country i .

$$football\ success\ share_i = \frac{football\ success_i}{\sum_j football\ success_j} = \frac{population_i}{\sum_j population_j} = popshare_i \quad (1)$$

Gásquez and Royuela (2016) point out some differences between the Olympic Games and international football which weaken the model. Firstly, in football 11 players are sent out onto the pitch regardless of the country's size, at the Qatar WC for example squad sizes were 26 men for each country (Bonn, 2022). In addition, the measure of "success" is very different, in the Olympics country's simply win medals based on positional performance, in international football higher performance equates to increased FIFA rankings or in this papers case, a higher Elo rating. Since the Elo rating has a weighting system, winning matches at the final stages of the WC gives more points than winning a regional match and therefore there is inequality in how the Elo is distributed, this is not a weakness of the Elo system and rather gives the need to modify the model to suit international football.

Natural talent in football requires technical features, expenditure on commodities, personnel and equipment are important in developing talent. Since wealthier countries are more likely to enable investment through private and public organisations, socioeconomic factors related to wealth must be included in the model (Gásquez and Royuela, 2016). A temperature that isn't overly hot or cold is also required to nurture popularity of the sport. In addition, Leeds and Leeds (2009) argue that the quality of national institutions lends itself to the political regime which they are under, better quality institutions can provide facilities and coaching. As an original addition to this, conflict involvement is jointly considered alongside regime considering its impact on resource allocation, which affects the quality of football institutions and the ability to capitalise on resources and labour. Lastly, cultural and traditional factors are key to success, Fan et al (2023) comments that the experience of hosting a World Cup enhances national performance, supporting the idea of a "cultural affinity towards football" (Leeds and Leeds, 2009:372). Combining these factors and building upon models by Bernard and Busse (2004), Gásquez and Royuela (2016) and Leeds and Leeds (2009), a production function for international football success can be specified where football success for country i in year t requires a population, N , wealth and resources, Y , a moderate temperature, T , quality football institutions, I and the ability to mobilize resources and labour, A .

$$football\ success_{i,t} = f(N_{i,t}, Y_{i,t}, T_{i,t}, I_{i,t}, A_{i,t}) \quad (2)$$

3.4. Model and Method

This paper examines the socioeconomic determinants of international football performance and their significance. With a particular focus on the impact of development and standard of living, as well as a political viewpoint. Using OLS and negative binomial regression this study estimates a panel data of 147 countries between 1994 and 2018. Building on the production function we can specify our first model:

$$ELO_{it} = \beta_0 + \beta_1 HDI_{it} + \beta_2 LPOP_{it} + \beta_3 TEMP14_{it} + \beta_4 DEMOC_{it} + \beta_5 CONFLICT_{it} \\ + \beta_6 PASTHOST_{it} + \beta_7 MEMBER_{it} + \beta_8 HOST_{it} + \varepsilon_{it}$$

We also include an additional model which features an interaction term. The model captures the interaction between HDI and political regime, allowing us to make inferences on the impacts of wealth and development, given a certain level of political freedom and regime type. Model two:

$$ELO_{it} = \beta_0 + \beta_1 (HDI * DEMOC)_{it} + \beta_2 LPOP_{it} + \beta_3 TEMP14_{it} + \beta_4 CONFLICT_{it} \\ + \beta_5 PASTHOST_{it} + \beta_6 MEMBER_{it} + \beta_7 HOST_{it} + \varepsilon_{it}$$

Generally speaking, there is a linear relationship between socioeconomic indicators like wealth, development, political stability and football performance. The majority of literature employs ordinary least squares (OLS) methods to estimate the effect of these determinants of performance, reflected through simplicity, interpretability and familiarity (Hoffmann, Ging and Ramasamy, 2002; Torgler, 2006; Leeds and Leeds 2009; Jiménez, 2016). Leeds and Leeds (2009) and Berlinschi, Schokkaert and Swinnen (2010) use a negative binomial regression as a robustness check, using a count variable such as FIFA ranking. In this study, we use the Elo rank to reflect a

Poisson distribution, generalising the negative binomial regression by loosening the assumption that the variance is equal to the mean made by the model (Cameron and Trivedi, 1998). While the negative binomial regression loses a degree of information regarding the difference in position between two nations, it allows us to check for robustness, assessing the sensitivity of findings and to model overdispersion. We can define model 3 as:

$$RANK_{it} = \beta_0 + \beta_1 HDI_{it} + \beta_2 LPOP_{it} + \beta_3 TEMP14_{it} + \beta_4 DEMOC_{it} + \beta_5 CONFLICT_{it} + \beta_6 PASTHOST_{it} + \beta_7 MEMBER_{it} + \beta_8 HOST_{it} + \varepsilon_{it}$$

In the models, β_0 represents the constant / intercept. i represents the country-specific effects, t represents the time effects. ε_{it} is the error-term of the model, in terms of country and time-specific effects.

4. Analysis

4.1. Diagnostic Tests

This paper examines the socioeconomic determinants of international football performance and their significance. With a particular focus on the impact of development and standard of living, as well as a political viewpoint. To pick the most appropriate model and build an accurate specification which reflects both the existing literature and the update on the study, diagnostic tests will be carried out to test for functional form, heteroscedasticity and overdispersion.

The Ramsey RESET test detects issues of incorrect functional form and misspecification which are commonplace in regression analysis (Ramsey, 1969; Wooldridge, 2012). The test's null hypothesis states that the model has no omitted variables, indicating that the functional form is correct. The F-statistic of 0.65 is not significant at the 1%, 5% and 10% level, therefore we have no evidence to reject the null hypothesis and conclude that the model is of the correct functional form, with no omitted variables (see table 3 in appendix).

Multicollinearity is a high correlation between two or more independent variables, and while it doesn't violate any of Gauss Markov's OLS assumptions, it reduces the

efficiency of the inference. This diminishing the statistical power of the model and makes it difficult to detect significant relationships between the dependent and independent variables (Wooldridge, 2012). We can use a Variance Inflation Factor (VIF) test to detect multicollinearity within the independent variables, all of the VIF values fall within the acceptable range (table 4) and therefore we can conclude that there are no issues of multicollinearity in the model.

The presence of heteroscedasticity will invalidate standard inference on our results, therefore the model is tested for non-constant variance. Using the Breusch-Pagan test for heteroscedasticity (table 5), we assume that the errors are distributed normally, the chi2 statistic of 0.0000 is significant at all levels, leading us to reject the null hypothesis and conclude that there is heteroscedasticity within the model (Wooldridge, 2012). We can support the conclusion by utilising White's test (table 6), which also gives us a significant p-value of 0.0000. Following Leeds and Leeds (2009), where there is heteroscedasticity of unknown form, we can use robust standard errors to account for the differing variances in the residuals.

Through the negative binomial regression, we can perform a likelihood ratio test to compare the suitability of the model to a Poisson model. The chi squared value of 4.8e+04 and the associated p -value of 0.000 suggests that alpha is non-zero and the negative binomial model is more appropriate than the Poisson model (Table 7).

4.2. Results and Discussion

Table 8 – Regression Results for Model 1 (OLS)

Variable	Coefficient	t-value	p-value
HDI	0.2432	10.52	0.000
LPOP	0.0091	3.61***	0.000
TEMP14	-0.0093	-12.07***	0.000
CONFLICT	-0.0241	-2.21**	0.027
DEMOC	0.0007	1.14	0.254
PASTHOST	0.1134	10.49***	0.000
MEMBER	0.0008	5.78***	0.000
HOST	0.0387	1.80*	0.071
$R^2 = 0.2735$			
Note: Dependent variable is Elo rating			
* , ** , *** indicate significance at the 10%, 5% and 1% respectively			

Results from the OLS regression can be seen in table 8 above. HDI is statistically significant at the 1% level, with a positive impact on international football performance, validating our first hypothesis (H_1). Findings are not consistent with Fan et al (2023) who found that HDI was negatively related to World Cup performance. This is also inconsistent with Macmillan and Smith (2007) who used HDI as an alternative proxy for the ability to invest in sporting resources, with findings concluding a lack of statistical significance. Our findings suggest that increased life expectancy and educational standards facilitate the development of youth talent, alongside greater levels of wealth, countries with more investment in public programs, facilities and coaching can nurture talent to allow for better playing standards nationally. Since HDI enlarges people's choice, opportunities arise for countries with higher levels of development, younger people can explore football as a career and not just a hobby, producing more talent at a stable rate (Lind, 1991). Results support Gásquez and Royuela (2014)'s paper on the inverse relationship, where it's mentioned that physical inactivity is a modifiable risk factor for a number of diseases. More important though, investment and wealth are not an isolated as an explanatory variable for football performance, in the sense that it's not simply the quality of institutions, coaching, facilities but also the quality of life, happiness and

level of health. Unlike Hoffmann et al (2002b:265)'s conclusion that football performance and wealth have an "inverted U-shape", our findings suggest that there are increasing returns for greater levels of HDI when measured against Elo rating. Figure 1 shows the relationship between international football performance and HDI, with increases in Elo rating gradually speeding up as development rises.

Population does, however, form a quadratic relationship, which is subject to the law of diminishing returns (figure 2). Significant at the 1% level, the optimum population is found to be approximately 550 million, any increases after that are subject to worse levels of international football performance, highlighting countries like the USA, India and China, who are all relatively poor compared to much smaller countries like France, Spain and Italy.

Deviations from 14 degrees Celsius resulted in worse overall performances in all football competitions, supporting Hoffmann, Geng and Ramasamy (2002b). The p -value is statistically significant, and the negative coefficient implies that there is a quadratic relationship between climate and performance. Figure 3 shows the relationship between temperature and Elo rating, interestingly the optimum temperature appears to shift downwards towards 10-11 degrees Celsius. This implies that a slightly cooler temperature is optimum when all international matches are considered, and not just the WC. As the game has modernised so have the facilities around the world, indoor pitches are more commonplace and youth talent can be nurtured without the worry of temperature, whether hot or cold.

Democracy and autocracy are not statistically significant within the regression, with a p -value of 0.244 and a very slight positive coefficient. The start of the panel data in the mid-1990s corresponds with a collapse in famously autocratic regimes, the Soviet Union and Yugoslavia, for example. Because there are less autocratic regimes in the 21st century, it's logical that regime type wouldn't have a significant impact on footballing performance, especially as regime does not specifically inhibit a nation's economic or productive output. Our findings are somewhat consistent with Jiménez (2016), who finds a weak but statistically significant positive relationship between democratic scale and winning % in all international competitions. A small coefficient of 0.0023 suggests that in terms of sports, autocratic and democratic regimes are increasingly indistinguishable, both can provide the facilities and

programs to develop talent to a professional level, regardless of freedom or regime. Findings do not support our second hypothesis regarding political regime (H_2), illustrating that for future literature, a different proxy or combination of political variables may be needed.

Interestingly, our conflict variable may prove to be more effective in regard to political and socioeconomic stability, as a negative relationship was found to be statistically significant at the 5% level, unlike democracy this is not consistent with Jiménez (2016) who found a statistically insignificant relationship using a stability variable. This may be a systematic difference in data, where we used a more accurate proxy of instability and violence in the 21st century, where major political shifts are not commonplace and therefore conflict is the most accurate indicator of instability. Findings from this paper suggest that conflict significantly pulls the resources and attention away from governments, resulting in lower sporting expenditure and less political encouragement for the public to pursue professional sports. In some cases, especially in more autocratic countries with mandatory military service, aspiring and promising athletes may have been pulled away from sports to represent their country in a different way; through conflict and fighting. These findings support our third hypotheses (H_3).

Considering the impact of football tradition and culture, we can explore the results of the PASTHOST and MEMBER proxies. Both variables are statistically significant at the 1% level, and both have a positive effect on international performance. The impact of previously hosting a FIFA World Cup is greater than the length of association with FIFA, this makes sense as in the period of 1994-2018, only France and Spain won the tournament without winning it previously. Hosting the World Cup has the second strongest coefficient, suggesting that it's one of the most important explanatory variables, these countries have proved they have the financial capabilities and love of football to host the World Cup, now they must prove themselves at each tournament to show the world their love of the game.

We find that for the World Cup and regional tournaments, a home advantage is not statistically significant, suggesting that nations elude the negative effects of playing in a foreign country or continent. This may reflect modern technology, the mindset of players or the incentives to play well regardless of geographical conditions. In

addition, due to regulations and transparency our findings suggest that referees are fair for both home and away teams, this may be because when team X plays team Y in an international competition, the referee has to be from neither country, regardless of the host nation.

4.3. Political-Economic Synergies

We included an additional model to model the interaction between human development levels and political regime. While democracy or autocracy are not statistically significant in our linear OLS regression, when we use an interaction term HDI is statistically significant when dependent on levels of democracy. This indicates that for the more democratic countries in the analysis, HDI has less of an impact with regards to international football performance. This goes against findings from Wan et al (2019) who found that a developed economy effectively enhances national football ability, especially in democracies. We conclude the opposite, that developed countries fare better in international tournaments given they are more autocratic than democratic. Perhaps this suggests that the most democratic countries focus their resources on non-sporting development, prioritising investment into the overall standard of living for citizens, rather than a focus on sports infrastructure. In less democratic countries, sports may be an outlook to facilitate national pride and patriotism.

4.4. Robustness Check

In line with Leeds and Leeds (2009), we can confirm the robustness of our specification through the use of a negative binomial regression, where we swap Elo Rating out for Elo Rank, a count variable. From table 10, all variables except host are statistically significant, with Democracy becoming significant in the negative binomial compared to the linear OLS, indicating that the influence of a democratic government on international football performance is more evident when exploring relative team position rather than overall points score. This could provide strong evidence for any future literature who wishes to explore the further impacts of democratic governance on international football performance, using a count variable.

5. Conclusion

In this paper we have analysed the socioeconomic and political determinants of performance in international football. The theoretical and empirical review facilitated debate and hypothesis and allowed us to formulate a logical theoretical framework, based on Bernard and Busse (2004) with political and institutional assistance from Leeds and Leeds (2009).

HDI appears to be a strong variable in measuring standard of living, wealth and development. While the more traditional GDP per capita is proved to be a valuable measurement in terms of football performance at a national level, it fails to account for factors relating to health and education, with the former clearly related to professional sports. Despite its critics, this study has proved that HDI is a sufficient alternative proxy to wealth, with the added benefit of accounting for more indicators of human development. In addition, in line with Gásquez and Royuela (2016), we conclude that the Elo rating is a better indicator than the FIFA rankings, the accuracy and span of the data is more suitable for panel data analyses.

Mixed results arise from the exploration of the Polity2 democratic index and a conflict dummy. Democratic nations only see their success in international football rise if they exist in a lower HDI bracket, that is that less developed democratic countries can perform well. There is evidence that, given a country is not fully democratic and developed, they can outperform fully democratic countries of any standard of living. Involvement in conflict supported our hypotheses and was statistically significant, this may provoke further exploration in future literature as the data is widely available. Nations involved in global or domestic warfare struggle to perform internationally. The impact of temperature on football performance follows previous literature, however the “optimal” temperature of 14 degrees Celsius is debated, with findings from this paper suggesting that in the modern game, this shifts downwards to reflect better standards of facilities (Hoffmann, Ging and Ramasamy, 2002b).

Countries with football ingrained in their culture and tradition exhibit higher average performances across all international competitions, which may suggest that nations new to the sport at the top level may struggle, future literature will pose the best option to explore this further as the last “new” team to win the World Cup was Spain

in 2010. A home advantage is found to not be statistically significant; a future study may opt to use another variable which explores “home continent effect.”

5.1. Future Research and Recommendations

To explore the political impacts, it's recommended that a greater number of variables are employed to account for different regimes, stability and economic freedom.

Dummy variables may be the right way to go here, as they are easily accounted for in terms of panel data, and they mitigate issues of multicollinearity. In modern times it's not simply sufficient to label a nation as X regime, data on government strength, public opinion and economic stability may provide a richer picture of a nation's political makeup.

It's recommended that Elo ratings are explored further, the only systematic negative to using the variable is the difficulty in gathering data, as a simple dataset does not exist online. FIFA rankings have been reputed across studies, with authors opting to use other variables like total points or goal difference (Fan et al, 2023; Scelles and Andreff, 2019). We advocate the use of Elo rating as an accurate indicator of international football performance, without limiting a study to only the World Cup.

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Appendix

Table 3 – Ramsey RESET Test

F(3, 3549)	0.54
P-Value	0.6528

Table 4 – VIF test for multicollinearity

VARIABLE	VIF	1/VIF
HDI	2.03	0.49
MEMBER	1.94	0.52
LPOP	1.51	0.66
TEMP14	1.50	0.66
PASTHOST	1.36	0.74
DEMOC	1.35	0.74
CONFLICT	1.27	0.79
HOST	1.04	0.96
MEAN	1.50	

Table 5 – Breusch-Pagan Test for Heteroscedasticity

Chi squared test statistic	52.09
<i>p-value</i>	0.0000

Table 6 – White's Test for Heteroscedasticity

Chi squared test statistic	291.43
<i>p-value</i>	0.0000

Table 7 – Likelihood Ratio Test

Chi bar test statistic	4.8e+04
<i>p-value</i>	0.0000

Table 9 – Regression results with interaction term (HDI x DEMOC)

Variable	Coefficient	t-value	<i>p-value</i>
LPOP	0.0082	3.19	0.001
TEMP14	-0.0098	-12.87	0.000
HDI	0.2553	10.32	0.000
DEMOC	0.0055	2.06	0.040
HDI x DEMOC	-0.0071	-2.00**	0.046
CONFLICT	-0.0224	-2.02	0.043
PASTHOST	0.1163	10.42	0.000

MEMBER	0.0009	6.59	0.000
HOST	0.0394	1.85	0.065
$R^2 = 0.2743$			
Note: Dependent variable is Elo rating			
* , ** , *** indicate significance at the 10%, 5% and 1% respectively			

Table 10 – Regression results from negative binomial regression

Variable	Coefficient	t-value	p-value
HDI	-0.5642	-8.92***	0.000
LPOP	-0.0486	-6.96***	0.000
TEMP14	0.0317	14.53***	0.000
CONFLICT	0.1953	5.35***	0.000
DEMOC	-0.0037	2.70***	0.007
PASTHOST	-0.8376	-11.99***	0.000
MEMBER	-0.0040	-8.24***	0.000
HOST	-0.1243	-1.13	0.258
$Wald\ Chi^2 = 0.000$			
Note: Dependent variable is Elo rating			
* , ** , *** indicate significance at the 10%, 5% and 1% respectively			

Figure 1 – Relationship between Elo Rating and HDI

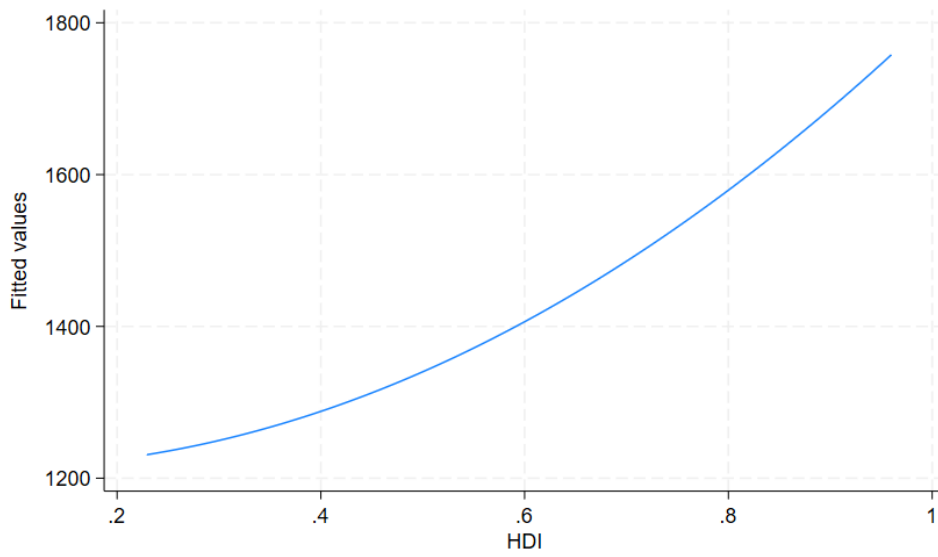


Figure 2 – Relationship between Elo Rating and Population

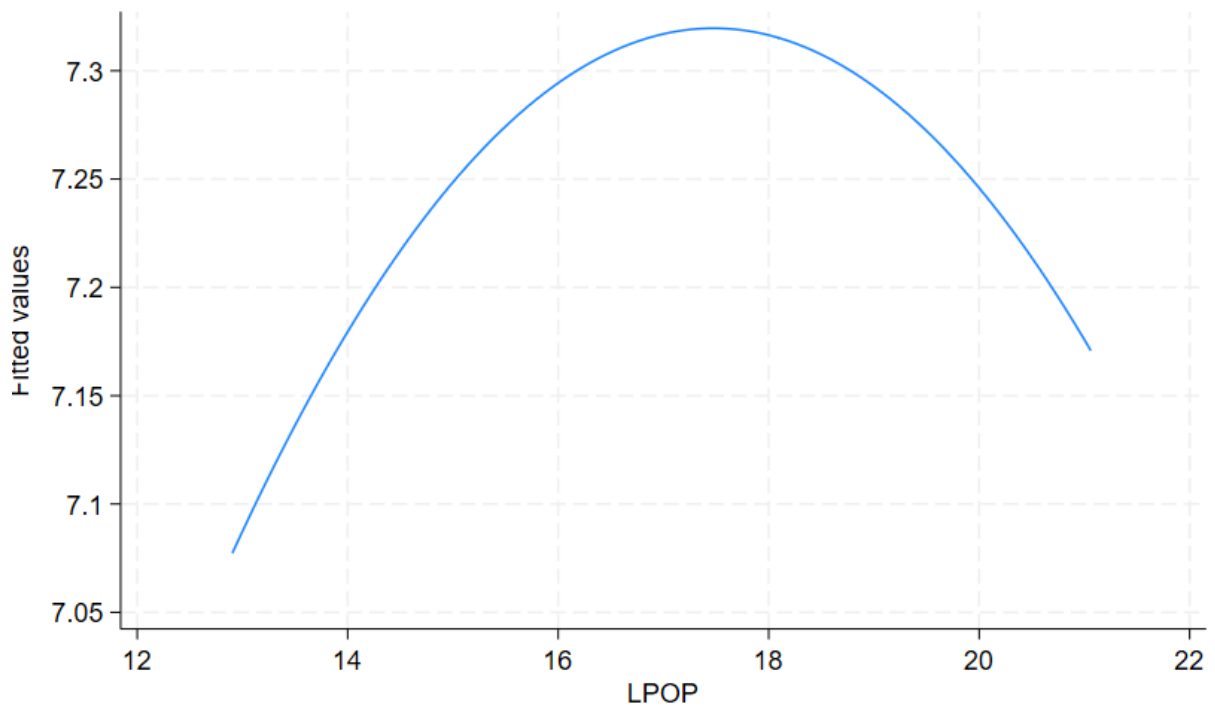


Figure 3 – Relationship between Elo Rating and Temperature

